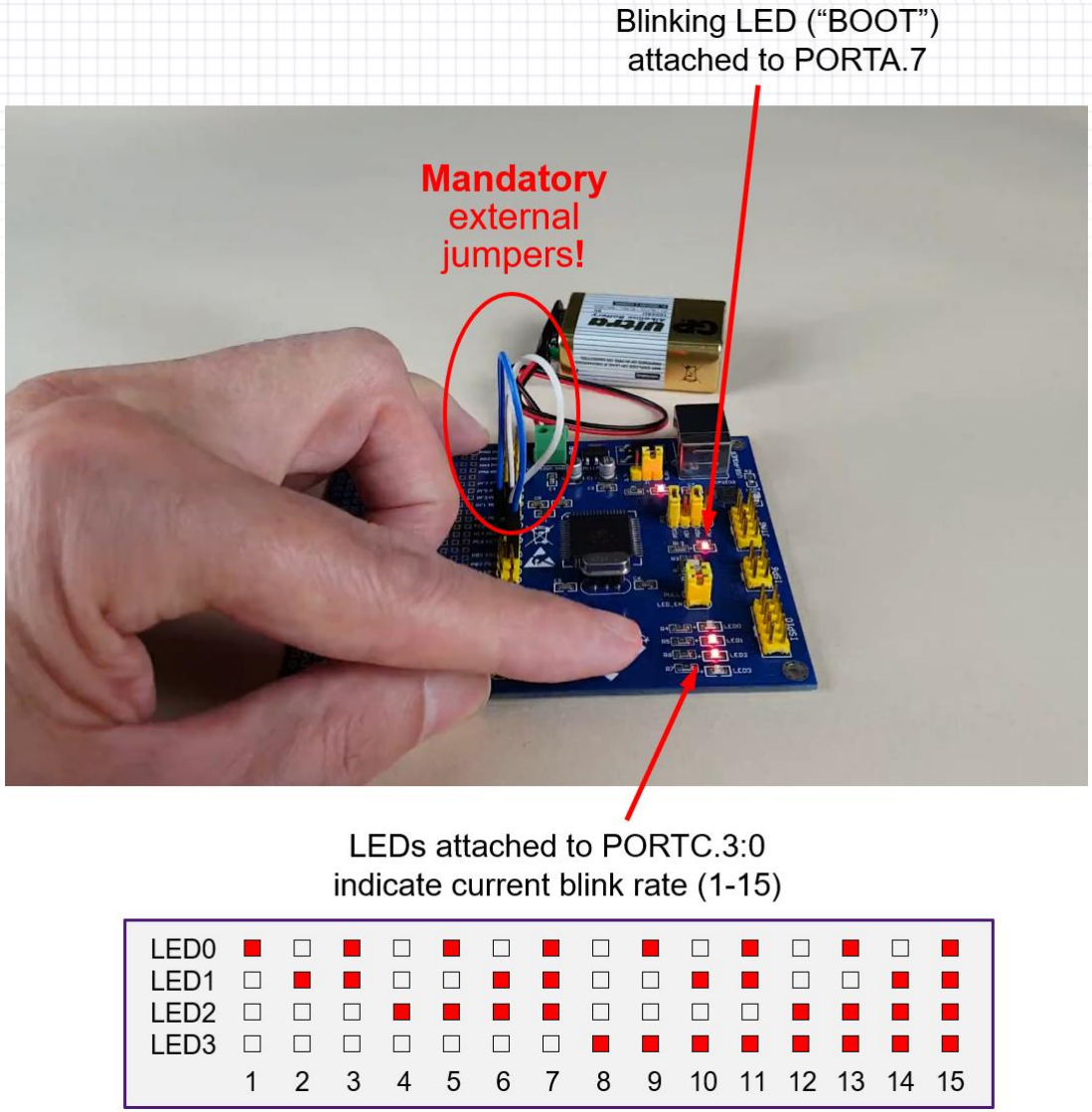
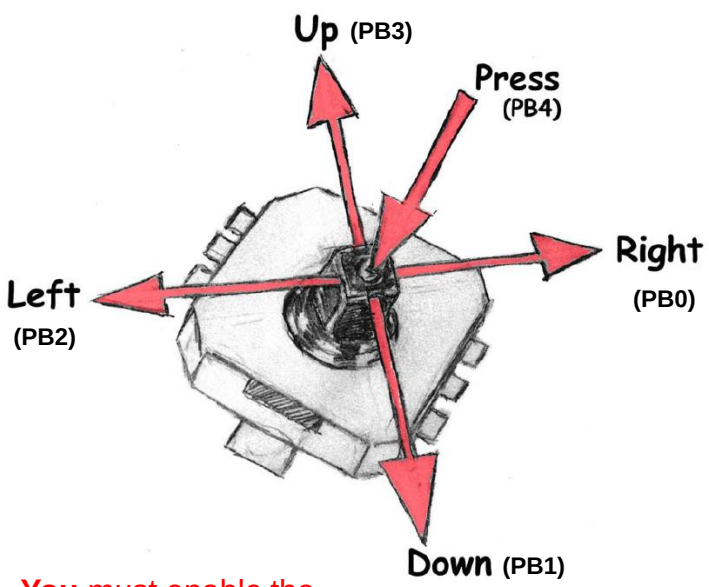
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Lab05 Notes

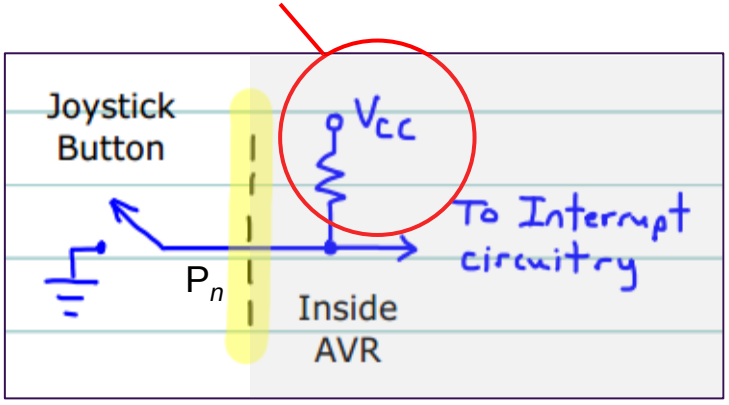
- Use ReadyAVR's joystick to control the blink rate of the "BOOT" LED
- Modify Lab02 to:
  - Blink the "BOOT" LED (**PORTA.7**) at a rate of 1-15Hz, as governed by a value stored in R20.
  - Use ISRs to change the value of R20 in response to up/down movement of the joystick.
  - Show current blink rate as a 4-bit binary number using LEDs connected to **PORTC.3:0**. Only update these LEDs when the blink rate is changed (i.e. in the ISR).

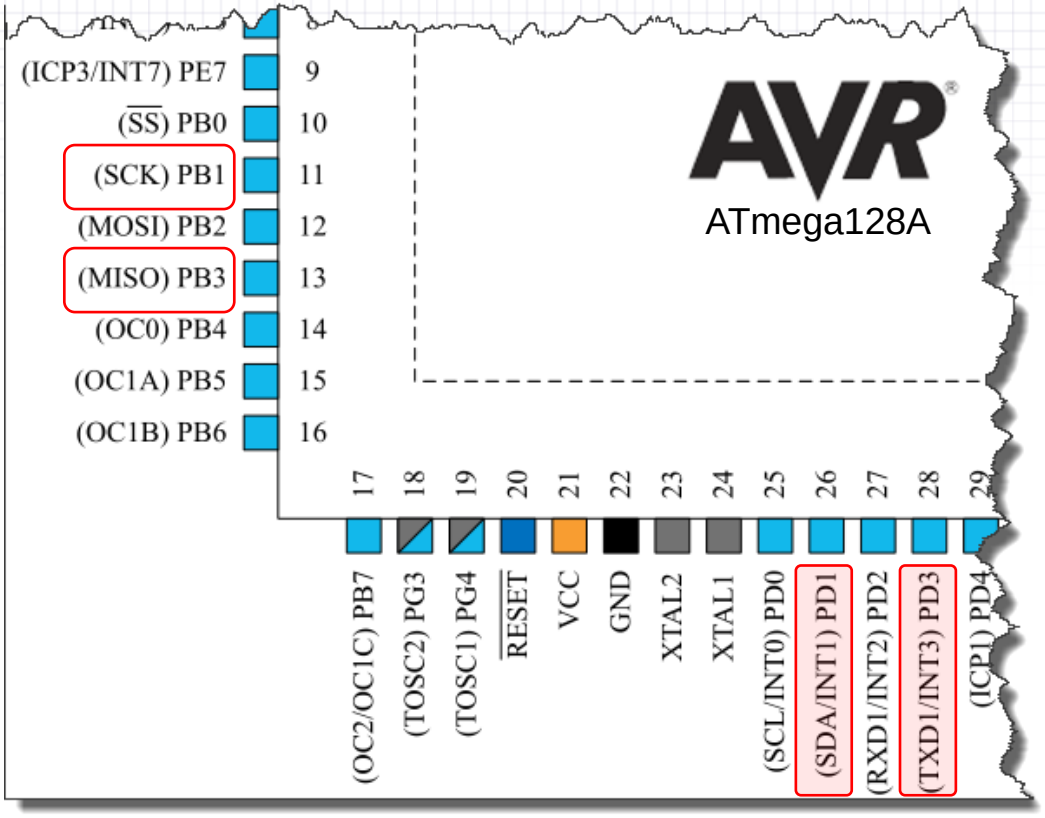


Lab05 Notes



You must enable the internal pull-up resistors





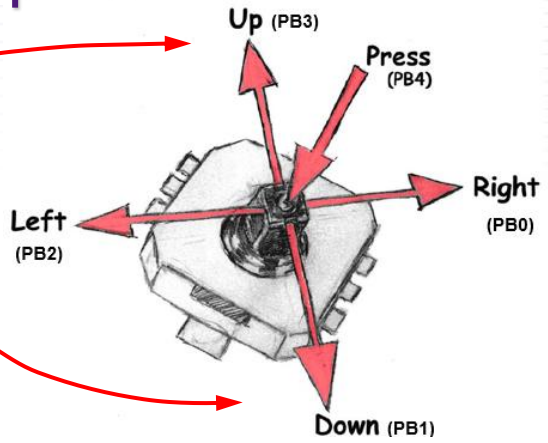
NB: Joystick switches are physically connected to PORTB pins, but those pins can't generate interrupts. However, PORTD pins can, and they can be connected **externally** to PORTB pins allowing joystick movements to invoke ISRs (see next page for details).

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Lab05 Notes

- As shown on the previous page, the ReadyAVR joystick is connect to five ATmega128A pins:

PORTB.4
PORTB.3
PORTB.2
PORTB.1
PORTB.0



- Key points:
  - PORTB.1 and PORTB.3 cannot generate interrupts
  - However, PORTD.1 and PORTD.3 ***can*** generate interrupts, so externally connect (via jumpers) PB1⇔PD1 and PB3⇔PD3
  - Then program PD1 (down) and PD3 (up) to trigger ISRs
  - MAKE SURE BOTH SETS OF PINS ARE CONFIGURED AS INPUTS!

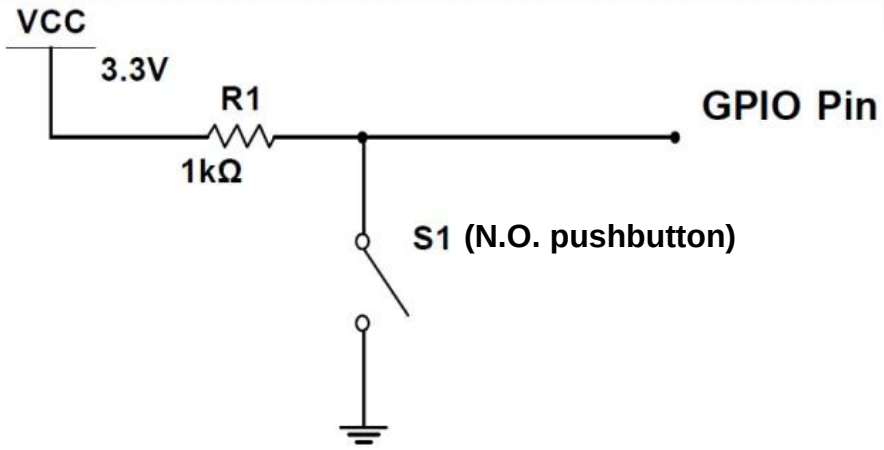
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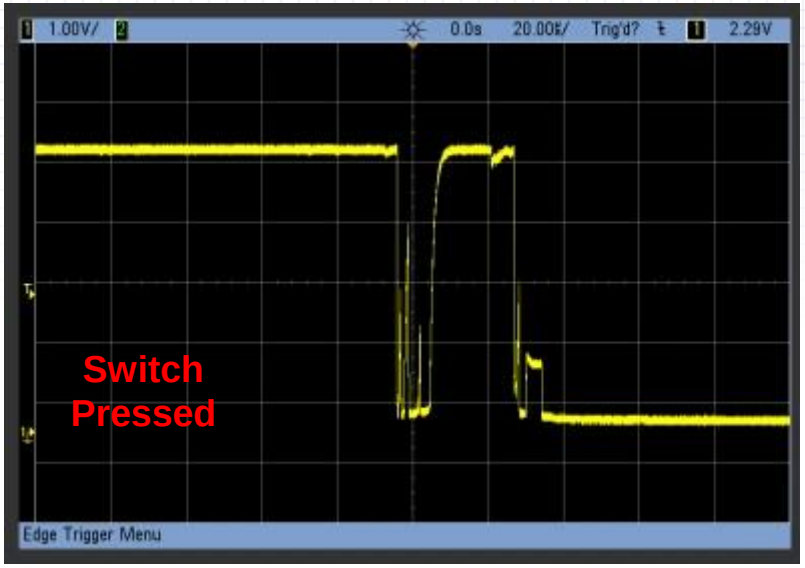
Lab05 Notes

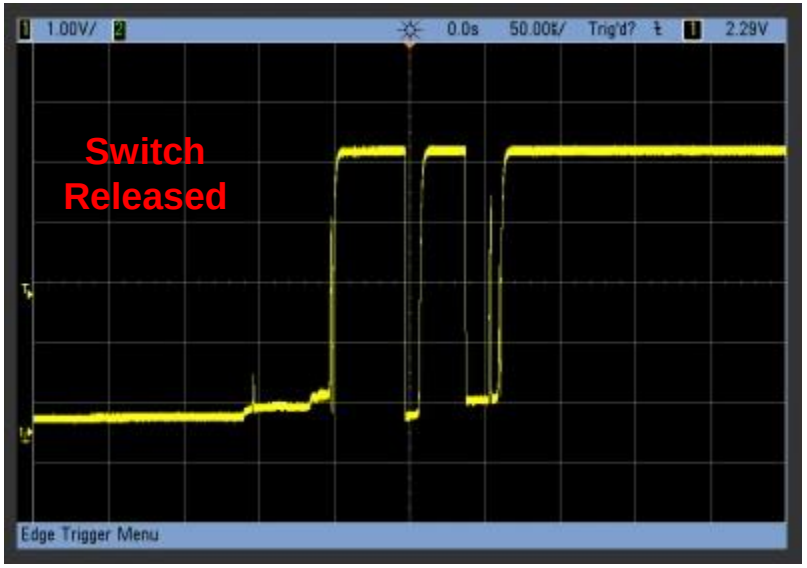
- You will need to configure several port- and interrupt-related registers for this lab:
  - Set INT1, INT3 ISR vectors (to handle up/down)
  - DDRA: set pin 7 as an output (the “BOOT” LED)
  - DDRB: set pins 1 and 3 as inputs (up/down joystick)
  - DDRC: set pins 3:0 as outputs (blink rate LEDs)
  - DDRD: set pins 1 and 3 as inputs (to trigger ISRs)
  - PORTB: set internal pull-ups on pins 1 and 3
  - EICRA: set INT1 and INT3 to activate on RISING edge
  - EIMSK: allow INT1 and INT3 to generate interrupts
  - SREG: enable interrupts globally (set bit 7 via sei)
- Read the ATmega128A data sheet slowly, carefully, and for full understanding!

Switch Bounce

Simple active low switch circuit



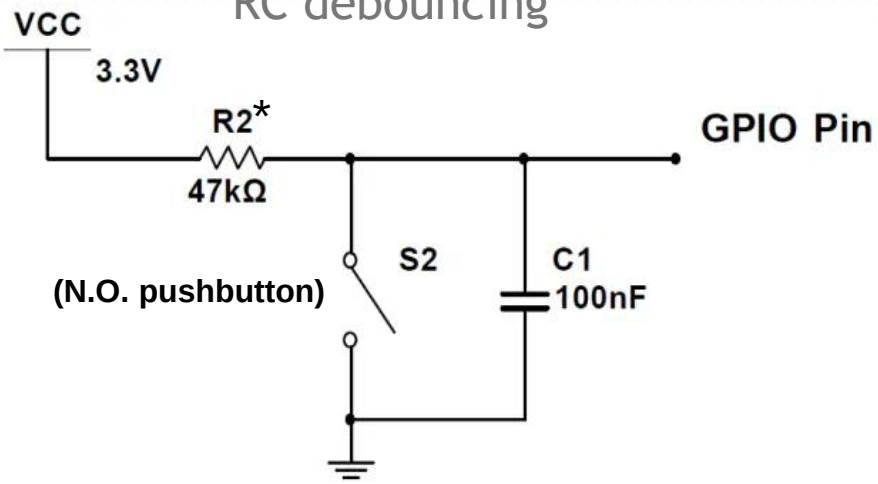


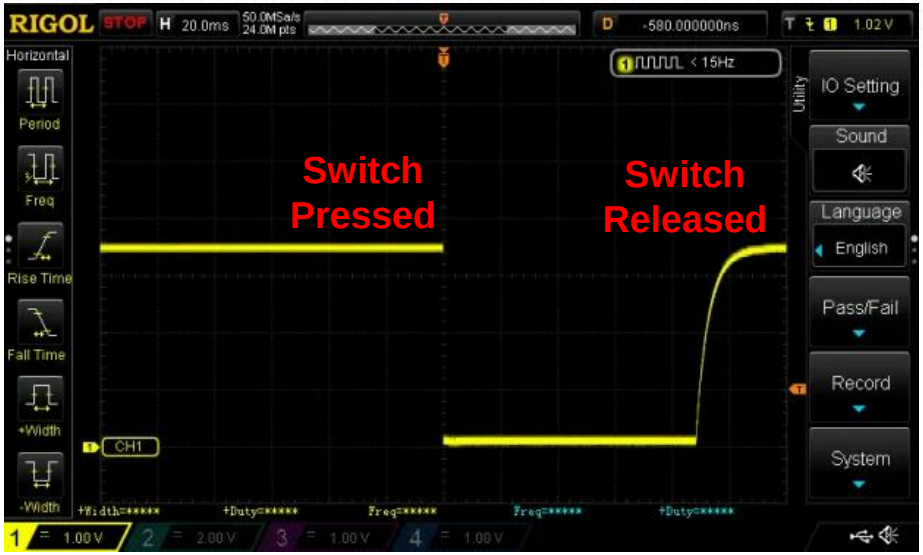


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RC Switch Debouncing

Simple active low switch circuit with RC debouncing





$$
\begin{aligned}
5\tau &= 5(R \times C) \\
&= 5(47k\Omega \times 100nF) \\
&= 5(4.7mS) \\
&= 23.5mS \approx \mathbf{25mS}
\end{aligned}
$$

*You can use the ATmega128A's internal pull-up resistor as R2, but you'll need to check the datasheet to discover it's value.